Hongbing Li

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RESEARCH INTERESTS

Music information retrieval, with interests in automatic music transcription (particularly performance-MIDI-to-score transcription), symbolic music modelling, and computational analysis of musical performance.

EDUCATION

Queen Mary University of London 09/2026 – Present
Master of Science in Sound and Music Computing Expected 08/2027

University of California, Santa Cruz 09/2018 – 07/2019
Master of Science in Statistics (Applied Economics), GPA: 3.81/4.00

Jinan University, China 09/2014 – 07/2018
Bachelor of Engineering in Computer Science (Information Security)

MANUSCRIPTS

Hongbing Li and Yaolong Ju (2026). *Structural Context from Audio and Lyric Evidence for Music Structure Analysis Using the New SheetSage-MSA Dataset* (under review at ICASSP 2027). arXiv:2610.26253.

RESEARCH EXPERIENCE

Accurate Music Structure Analysis with Audio and Lyrics

- Developed a model with separate components for detecting section changes and labeling verses, choruses and other sections; used repeated lyrics to improve chorus identification.
- Released SheetSage-MSA with 13,880 annotated excerpts from 9,010 songs (82.5 hours of annotated audio), enabling training on partially annotated recordings.
- Achieved state-of-the-art results on a public 200-song test set, improving boundary F1 over the previous best system by 3.5 and 4.2 percentage points at 0.5 s and 3 s tolerances, and section-label accuracy by 0.9 points.

Performance MIDI-to-Score Transcription with Limited Paired Data *Ongoing*

- Explored unpaired training with score-only PDMX data, and investigated methods for turning weakly paired PianoCoRe performances and scores into aligned training pairs.
- Designed a new transcription architecture incorporating auxiliary rhythm prediction; initial experiments showed lower note-duration error.
- Conducted case studies to identify transcription error patterns and limitations of existing evaluation metrics, guiding model development and ongoing metric refinement.

INDUSTRY EXPERIENCE

Mindigital Fintech Co., Ltd. 04/2022 – 06/2026
Data Scientist

- Built machine learning pipelines spanning training, deployment and monitoring; engineered features from heterogeneous data.

Meituan 10/2020 – 03/2022
Data Scientist

- Identified business growth opportunities using machine learning, deep learning and data analysis; communicated findings through reports and dashboards.

TECHNICAL AND MUSICAL BACKGROUND

Programming and methods: Python, PyTorch, librosa; sequence modeling, audio signal processing, multimodal learning.

Musical training: 10 years of piano; 3+ years of guitar.

Languages: English, Mandarin.